TUTORIAL 12

Inner Products, Orthogonality, and Projections

MOTIVATION AND GUIDING IDEA

An inner product adds geometric measurement to a vector space: we can ask whether two directions are perpendicular, measure size, and decompose a vector relative to a subspace. A different inner product changes the geometry even when the vectors and subspace remain fixed. Begin every computation by identifying the given inner product, and only then apply the formulas.

NOTATION AND CONVENTIONS

- $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ is the scalar field; $\overline{z}$ is complex conjugation, with $\overline{z} = z$ over the reals.
- A^T is the transpose and $A^* = \overline{A}^T$ is the conjugate transpose. For column vectors, the standard inner product is $\langle x, y \rangle = x^T \overline{y} = y^* x$.
- $\langle v, w \rangle$ denotes the given inner product, and $\|v\| = \sqrt{\langle v, v \rangle}$ is its induced norm. For a weighted inner product, we sometimes write $\langle v, w \rangle_G$ and $\|v\|_G$.
- $v \perp w$ means $\langle v, w \rangle = 0$; $S^\perp$ is the set of all vectors orthogonal to every element of S.
- δ_{ij} is the Kronecker delta: it is 1 when $i = j$ and 0 otherwise. Thus $\langle e_i, e_j \rangle = \delta_{ij}$ expresses orthonormality.
- P_W is the orthogonal projection map onto W; $P_W(v)$ is the projection of v onto W, and $v - P_W(v)$ is the orthogonal residual.
- $V = W \oplus W^\perp$ means that every $v \in V$ has a unique decomposition $v = w + z$ with $w \in W$ and $z \in W^\perp$.
- $[v]_B$ is the coordinate vector of v relative to the **ordered** basis B; the basis order determines the coordinate order.
- $G \succ 0$ and $G \succeq 0$ denote a positive definite Hermitian matrix and a positive semidefinite Hermitian matrix, respectively.

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DEFINITION 12.1

Let V be an inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. We use linearity in the first argument and conjugate linearity in the second. Thus, for $u \neq 0$, the projection coefficient onto $\operatorname{span}\{u\}$ is

$$\frac{\langle v, u\rangle}{\langle u, u\rangle},$$

with this order of the arguments in the numerator. The denominator is strictly positive. If V is finite dimensional and $W \leq V$, then $P_W(v)$ is uniquely characterized by the two conditions together:

$$P_W(v) \in W, \qquad v - P_W(v) \in W^{\perp}.$$

If $(e_1, \ldots, e_k)$ is an orthonormal basis of W, then $P_W(v) = \sum_{j=1}^{k} \langle v, e_j\rangle e_j$. For an orthogonal basis that is not normalized, divide each coefficient by $\|e_j\|^2$. Membership alone is insufficient, and orthogonality alone is also insufficient.

MOTIVATION AND GUIDING IDEA

We study how an inner product measures lengths and angles, how to choose an orthonormal basis, and how orthogonal projection decomposes a vector into a component in a subspace and a component perpendicular to it. We connect algebraic computation with geometry and compare different inner products.

12.1 Inner Product Spaces

RECALL AND TOOLS

Throughout this chapter, let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and let V be a vector space over $\mathbb{F}$. Write $\bar{\cdot} : \mathbb{F} \to \mathbb{F}$ for standard conjugation: that is, if $\mathbb{F} = \mathbb{C}$, then $\overline{a + ib} = a - ib$; if $\mathbb{F} = \mathbb{R}$, then $\overline{x} = x$ for every $x \in \mathbb{R}$.

RECALL AND TOOLS

Let V be a vector space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. An *inner product* on V is a map

$$\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$$

satisfying the following axioms for all $u, v, w \in V$ and all $\alpha, \beta \in \mathbb{F}$.

1. **Linearity in the first argument:**

$$\langle \alpha u + \beta v,\ w\rangle = \alpha\langle u, w\rangle + \beta\langle v, w\rangle.$$

2. **Conjugate symmetry (Hermitian symmetry):**

$$\langle u, v\rangle = \overline{\langle v, u\rangle}.$$

3. **Positive definiteness:**

$$\langle v, v\rangle \in \mathbb{R}, \qquad \langle v, v\rangle \geq 0, \qquad \langle v, v\rangle = 0 \iff v = 0.$$

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Formal observation: Linearity in the first argument and conjugate symmetry imply *conjugate linearity in the second argument*: for all $\alpha, \beta \in \mathbb{F}$ and all $u, v, w \in V$,

$$\langle u,\, \alpha v + \beta w \rangle = \overline{\alpha}\, \langle u, v \rangle + \overline{\beta}\, \langle u, w \rangle.$$

When $\mathbb{F} = \mathbb{R}$, this gives ordinary linearity in the second argument as well.

EXAMPLE 12.1

Standard examples of inner products:

- On $\mathbb{C}^n$ (column vectors $u, v \in \mathbb{C}^n$):

$$\langle u, v \rangle := u^T \overline{v}.$$

  On $\mathbb{R}^n$:

$$\langle u, v \rangle := u^T v.$$

- On $M_{m\times n}(\mathbb{C})$:

$$\langle A, B \rangle := \operatorname{tr}\!\left(A^T \overline{B}\right).$$

- On $\mathbb{R}_n[x]$ (or on $C([a,b], \mathbb{R})$), where $a < b$:

$$\langle p, q \rangle := \int_a^b p(x)\, q(x)\, dx.$$

DEFINITION 12.2

Let $(V, \langle \cdot, \cdot \rangle)$ be a vector space equipped with an inner product. The **norm** induced by the inner product is defined, for each $v \in V$, by

$$\|v\| := \sqrt{\langle v, v \rangle}.$$

Thus $\|v\| \in [0, \infty)$; in particular, $\|v\| = 0$ if and only if $v = 0$.

DEFINITION 12.3

Inner product space: If V is a vector space over $\mathbb{F}$ and $\langle \cdot, \cdot \rangle$ is an inner product on V, then the pair

$$\big(V, \langle \cdot, \cdot \rangle\big)$$

is called an **inner product space**.

PROPOSITION 12.1

Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. For all $n, m \in \mathbb{N}$, all $\alpha_1, \ldots, \alpha_n, \beta_1, \ldots, \beta_m \in \mathbb{F}$, and all $v_1, \ldots, v_n, u_1, \ldots, u_m \in V$,

$$\left\langle \sum_{i=1}^{n} \alpha_i v_i,\ \sum_{j=1}^{m} \beta_j u_j \right\rangle = \sum_{i=1}^{n} \sum_{j=1}^{m} \alpha_i\, \overline{\beta_j}\, \langle v_i, u_j \rangle.$$

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When $\mathbb{F} = \mathbb{R}$, we have $\overline{\beta_j} = \beta_j$, so the formula reduces to

$$\left\langle \sum_{i=1}^{n} \alpha_i v_i,\ \sum_{j=1}^{m} \beta_j u_j \right\rangle = \sum_{i=1}^{n} \sum_{j=1}^{m} \alpha_i\, \beta_j\, \langle v_i, u_j \rangle.$$

12.2 Basic Exercises and Results

EXERCISE 12.1

Let $n \geq 1$. On $V = M_n(\mathbb{R})$, define

$$f(A, B) := \mathrm{tr}(AB).$$

Please determine for which values of n this is an inner product, and

WORKED SOLUTION

If $n = 1$, identify matrices with real numbers. Then $f(a, b) = ab$ is the standard inner product on $\mathbb{R}$. If $n \geq 2$, choose $A = E_{12}$: that is, $a_{12} = 1$ and all other entries are zero. Then $A \neq 0$, but $A^2 = 0$, so

$$f(A, A) = \mathrm{tr}(A^2) = 0.$$

Positive definiteness therefore fails, and the function is not an inner product when $n \geq 2$.

EXERCISE 12.2

Let $n \in \mathbb{N}$, and let $M \in M_n(\mathbb{R})$ be symmetric and positive definite; that is,

$$M^T = M, \qquad x^T M x > 0 \ \ \forall x \in \mathbb{R}^n \setminus \{0\}.$$

Define $\langle \cdot, \cdot \rangle_M : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ by

$$\langle x, y \rangle_M := x^T M y.$$

1. Please prove that $\langle \cdot, \cdot \rangle_M$ is an inner product on $\mathbb{R}^n$.
2. Please show that the induced norm is $\|x\|_M = \sqrt{x^T M x}$.
3. Physical example: suppose a linear mechanical system is described by a mass matrix M (in some system of generalized coordinates). Please show that its kinetic energy

$$T(v) = \frac{1}{2}\, v^T M v$$

equals $\frac{1}{2}\|v\|_M^2$.

WORKED SOLUTION

1. We verify the inner product axioms.

   Linearity in the first argument: Let $x_1, x_2, y \in \mathbb{R}^n$ and $\alpha, \beta \in \mathbb{R}$. Then

$$\langle \alpha x_1 + \beta x_2,\ y \rangle_M = (\alpha x_1 + \beta x_2)^T M y = \alpha x_1^T M y + \beta x_2^T M y = \alpha \langle x_1, y \rangle_M + \beta \langle x_2, y \rangle_M.$$

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EXERCISE 12.2 CONTINUED

Symmetry: For every $x, y \in \mathbb{R}^n$,

$$\langle x, y \rangle_M = x^T M y.$$

Since this is a real scalar, $(x^T My)^T = x^T My$, so

$$\langle x, y \rangle_M = (x^T My)^T = y^T M^T x = y^T Mx = \langle y, x \rangle_M,$$

where we used $M^T = M$.

Positive definiteness: For every $x \in \mathbb{R}^n$,

$$\langle x, x \rangle_M = x^T Mx \geq 0,$$

and $\langle x, x \rangle_M = 0$ if and only if $x = 0$, because M is positive definite.

Hence $\langle \cdot, \cdot \rangle_M$ is an inner product on $\mathbb{R}^n$.

2. By the definition of the norm induced by an inner product, for every $x \in \mathbb{R}^n$,

$$\|x\|_M := \sqrt{\langle x, x \rangle_M} = \sqrt{x^T Mx}.$$

3. For every $v \in \mathbb{R}^n$,

$$T(v) = \frac{1}{2} v^T Mv = \frac{1}{2} \langle v, v \rangle_M = \frac{1}{2} \|v\|_M^2,$$

so the kinetic energy is half the square of the norm induced by $\langle \cdot, \cdot \rangle_M$.

EXERCISE 12.3

Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and $A \in M_n(\mathbb{F})$. For any matrix C, please write $C^* := \overline{C}^T$; over $\mathbb{R}$, $C^* = C^T$. Following the convention that the inner product is linear in its first argument, define

$$\langle x, y \rangle := y^* x = x^T \overline{y}.$$

1. Please prove that $\langle Av, w \rangle = \langle v, A^* w \rangle$ for all $v, w \in \mathbb{F}^n$.
2. Let $G \in M_n(\mathbb{F})$ be Hermitian and positive definite. Define

$$\langle x, y \rangle_G := y^* Gx.$$

Please prove that this is an inner product, and please show that there is a unique matrix A^{*G} satisfying

$$\langle Av, w \rangle_G = \langle v, A^{*G} w \rangle_G \quad \text{for all } v, w \in \mathbb{F}^n,$$

and that $A^{*G} = G^{-1} A^* G$.

WORKED SOLUTION

1. The identity $(A^* w)^* = w^* A$ gives

$$\langle Av, w \rangle = w^* Av = (A^* w)^* v = \langle v, A^* w \rangle.$$

2. Linearity in the first argument follows from the linearity of matrix multiplication. Since

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EXERCISE 12.3 CONTINUED

$G^* = G$,

$$\overline{\langle y, x\rangle_G} = \overline{x^*Gy} = (x^*Gy)^* = y^*G^*x = y^*Gx = \langle x, y\rangle_G.$$

Furthermore, $\langle x, x\rangle_G = x^*Gx > 0$ for every $x \neq 0$, and $\langle 0, 0\rangle_G = 0$. The matrix G is invertible: otherwise $Gx = 0$ for some $x \neq 0$ would imply $x^*Gx = 0$. Set $B := G^{-1}A^*G$. Then

$$B^* = (G^{-1}A^*G)^* = GAG^{-1},$$

and hence

$$\langle v, Bw\rangle_G = (Bw)^*Gv = w^*B^*Gv = w^*GAv = \langle Av, w\rangle_G.$$

For uniqueness, if C also satisfies the identity, then $w^*(C^*G - GA)v = 0$ for all v, w. Choosing $v = e_j$ and $w = e_i$ shows that every entry of $C^*G - GA$ is zero. Thus

$$C^*G = GA \implies C^* = GAG^{-1} \implies C = G^{-1}A^*G = B.$$

EXERCISE 12.4

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space over $\mathbb{C}$, and let $u, v \in V$ satisfy $u \neq v$. Please show that there is $w \in V$ such that $\langle w, u\rangle \neq \langle w, v\rangle$.

WORKED SOLUTION

Choose $w = u - v$. Since $u \neq v$, the vector w is nonzero. Conjugate linearity in the second argument implies additivity, so

$$\langle w, u\rangle - \langle w, v\rangle = \langle w, u - v\rangle = \langle u - v, u - v\rangle = \|u - v\|^2 > 0.$$

Therefore $\langle w, u\rangle \neq \langle w, v\rangle$, as required.

12.3 Orthogonality

DEFINITION 12.4

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space. Vectors $v, w \in V$ are **orthogonal** if $\langle v, w\rangle = 0$; we write $v \perp w$.

PROPOSITION 12.2

If $v \perp b$ for every $b \in B$, then $v \perp \operatorname{span}(B)$. Indeed, for any linear combination $s = \sum_{j=1}^{k} \alpha_j b_j$,

$$\langle v, s\rangle = \sum_{j=1}^{k} \overline{\alpha_j}\langle v, b_j\rangle = 0.$$

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DEFINITION 12.5

A set $B = \{v_1, \ldots, v_n\} \subseteq V$ is **orthogonal** if

$$\langle v_i, v_j \rangle = 0 \qquad \text{for all } i \neq j.$$

If, in addition, $\|v_i\| = 1$ for every i, it is **orthonormal**. In this case,

$$\langle v_i, v_j \rangle = \delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

If B is also a basis of V, we call it an orthogonal basis or an orthonormal basis, respectively.

RECALL AND TOOLS

Normalization: an orthogonal set can be converted into an orthonormal set: if $\{v_1, \ldots, v_n\}$ is orthogonal and contains no zero vector, then $\left\{ \frac{v_1}{\|v_1\|}, \ldots, \frac{v_n}{\|v_n\|} \right\}$ is orthonormal. Indeed, for every $1 \leq i \leq n$, $\left\langle \frac{v_i}{\|v_i\|}, \frac{v_i}{\|v_i\|} \right\rangle = \frac{1}{\|v_i\|^2} \langle v_i, v_i \rangle = 1$.

PROPOSITION 12.3

- An **orthogonal** set containing no zero vector is linearly independent.
- An **orthonormal** set is linearly independent. (Since each of its elements has norm 1, it contains no zero vector.)
- The converse fails: for example, $\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$ with the standard inner product is linearly independent but is neither orthogonal nor orthonormal.

EXERCISE 12.5

Let $u, v \in \mathbb{R}^n$ be such that $a = 3v - 2u$ and $b = u - v$ are orthonormal for the standard inner product. Please compute $\|u\|$, $\|v\|$, and $\langle u, v \rangle$, and please prove that u, v are linearly independent.

WORKED SOLUTION

First solve the two vector identities: from $b = u - v$, we obtain $u = v + b$. Substitution into $a = 3v - 2u$ gives $a = v - 2b$, so

$$v = a + 2b, \qquad u = a + 3b.$$

Since $\langle a, a \rangle = \langle b, b \rangle = 1$ and $\langle a, b \rangle = 0$,

$$\|v\|^2 = \langle a + 2b, a + 2b \rangle = 1 + 2 \cdot 0 + 2 \cdot 0 + 4 = 5,$$

$$\|u\|^2 = \langle a + 3b, a + 3b \rangle = 1 + 3 \cdot 0 + 3 \cdot 0 + 9 = 10,$$

$$\langle u, v \rangle = \langle a + 3b, a + 2b \rangle = 1 + 2 \cdot 0 + 3 \cdot 0 + 6 = 7.$$

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EXERCISE 12.5 CONTINUED

Thus $\|v\| = \sqrt{5}$, $\|u\| = \sqrt{10}$, and $\langle u, v\rangle = 7$. If $\alpha u + \beta v = 0$, then

$$(\alpha + \beta)a + (3\alpha + 2\beta)b = 0.$$

Since a, b are orthonormal, they are linearly independent. Hence $\alpha + \beta = 0$ and $3\alpha + 2\beta = 0$. Subtracting twice the first equation from the second gives $\alpha = 0$, and then $\beta = 0$. The given information therefore determines the norms and inner product and also establishes the linear independence of u, v.

12.4 Coordinates in an Orthogonal or Orthonormal Basis

DEFINITION 12.6

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Let $B = (b_1, \ldots, b_n)$ be an ordered basis of V.
Every $v \in V$ has a unique representation as a linear combination of the basis vectors: $v = \alpha_1 b_1 + \cdots + \alpha_n b_n$.
If B is an **orthogonal** basis, its coefficients are given by $\alpha_i = \frac{\langle v, b_i\rangle}{\|b_i\|^2}$.
If B is an **orthonormal** basis, then $\|b_i\|^2 = 1$, so $\alpha_i = \langle v, b_i\rangle$. Thus the coordinate vector relative to B is

$$[v]_B = \begin{pmatrix} \langle v, b_1\rangle \\ \vdots \\ \langle v, b_n\rangle \end{pmatrix}$$

EXERCISE 12.6

On $\mathbb{C}^3$, please use the inner product $\langle x, y\rangle = x^T\overline{y}$. Let

$$b_1 = \tfrac{1}{5}(3, 4i, 0)^T, \quad b_2 = \tfrac{1}{5}(4, -3i, 0)^T, \quad b_3 = (0, 0, 1)^T.$$

Please show that $B = (b_1, b_2, b_3)$ is an orthonormal basis and please find $[v]_B$ for $v = (1, 3i, 2)^T$.

WORKED SOLUTION

Compute the norms:

$$\|b_1\|^2 = \frac{3^2 + |4i|^2}{25} = \frac{9 + 16}{25} = 1, \quad \|b_2\|^2 = \frac{4^2 + |-3i|^2}{25} = 1, \quad \|b_3\|^2 = 1.$$

The only inner product requiring a calculation is

$$\langle b_1, b_2\rangle = \frac{3 \cdot 4 + (4i)\overline{-3i}}{25} = \frac{12 + (4i)(3i)}{25} = \frac{12 - 12}{25} = 0.$$

The two inner products with b_3 vanish because of the third coordinate. Thus B is orthonormal; since it has three vectors in $\mathbb{C}^3$, it is a basis. With our convention of linearity in the first

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EXERCISE 12.6 CONTINUED

argument, the coefficients are $\langle v, b_j \rangle$:

$$\langle v, b_1 \rangle = \frac{1 \cdot 3 + (3i)(-4i)}{5} = \frac{3 + 12}{5} = 3,$$

$$\langle v, b_2 \rangle = \frac{1 \cdot 4 + (3i)(3i)}{5} = \frac{4 - 9}{5} = -1, \qquad \langle v, b_3 \rangle = 2.$$

Hence $[v]_B = (3, -1, 2)^T$. Check by reconstruction:

$$3b_1 - b_2 + 2b_3 = \left(\frac{9-4}{5}, \frac{12i + 3i}{5}, 2 \right)^T = (1, 3i, 2)^T = v.$$

12.5 Norms and the Cauchy-Schwarz Inequality

DEFINITION 12.7

Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and let V be a vector space over $\mathbb{F}$. A **norm** on V is a function

$$\| \cdot \| : V \to [0, \infty)$$

satisfying the following axioms for all $u, v \in V$ and all $\alpha \in \mathbb{F}$:

1. **Positive definiteness:**
$$\|v\| \geq 0, \qquad \|v\| = 0 \iff v = 0.$$
2. **Homogeneity:**
$$\|\alpha v\| = |\alpha| \, \|v\|.$$
3. **Triangle inequality:**
$$\|u + v\| \leq \|u\| + \|v\|.$$

Here $|\alpha|$ denotes absolute value over $\mathbb{R}$ or modulus over $\mathbb{C}$. A vector space equipped with a norm is a **normed space**, denoted $(V, \| \cdot \|)$.

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PROPOSITION 12.4

Cauchy-Schwarz inequality. For all $u, v \in V$,

$$|\langle u, v \rangle| \leq \|u\| \, \|v\|.$$

Equality holds if and only if u, v are linearly dependent: that is, there is $\alpha \in \mathbb{F}$ such that $u = \alpha v$ or $v = \alpha u$.

PROPOSITION 12.5

Angle in a real space. If $\mathbb{F} = \mathbb{R}$ and $u, v \neq 0$, the number

$$\cos \zeta := \frac{\langle u, v \rangle}{\|u\| \|v\|}$$

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satisfies $\cos\zeta \in [-1,1]$. In particular, there is $\zeta \in [0,\pi]$ for which this equality holds.

EXERCISE 12.7

Let $\omega, \zeta, \xi > 0$. Please prove the inequality

$$\frac{\omega}{\zeta+\xi} + \frac{\zeta}{\xi+\omega} + \frac{\xi}{\omega+\zeta} \geq \frac{(\omega+\zeta+\xi)^2}{2(\omega\zeta+\zeta\xi+\xi\omega)}.$$

WORKED SOLUTION

Use the standard inner product on $\mathbb{R}^3$, namely $\langle x, y\rangle = x^T y$.
Define the vectors

$$x = \left(\sqrt{\frac{\omega}{\zeta+\xi}}, \sqrt{\frac{\zeta}{\xi+\omega}}, \sqrt{\frac{\xi}{\omega+\zeta}}\right), \qquad y = \left(\sqrt{\omega(\zeta+\xi)}, \sqrt{\zeta(\xi+\omega)}, \sqrt{\xi(\omega+\zeta)}\right).$$

Then

$$\langle x, y\rangle = \sqrt{\frac{\omega}{\zeta+\xi}}\sqrt{\omega(\zeta+\xi)} + \sqrt{\frac{\zeta}{\xi+\omega}}\sqrt{\zeta(\xi+\omega)} + \sqrt{\frac{\xi}{\omega+\zeta}}\sqrt{\xi(\omega+\zeta)} = \omega+\zeta+\xi.$$

Also,

$$\|x\|^2 = \langle x, x\rangle = \left(\sqrt{\frac{\omega}{\zeta+\xi}}\right)^2 + \left(\sqrt{\frac{\zeta}{\xi+\omega}}\right)^2 + \left(\sqrt{\frac{\xi}{\omega+\zeta}}\right)^2 = \frac{\omega}{\zeta+\xi} + \frac{\zeta}{\xi+\omega} + \frac{\xi}{\omega+\zeta},$$

and

$$\begin{aligned}\|y\|^2 &= \langle y, y\rangle \\ &= \left(\sqrt{\omega(\zeta+\xi)}\right)^2 + \left(\sqrt{\zeta(\xi+\omega)}\right)^2 + \left(\sqrt{\xi(\omega+\zeta)}\right)^2 \\ &= \omega(\zeta+\xi) + \zeta(\xi+\omega) + \xi(\omega+\zeta).\end{aligned}$$

Expand the last sum:

$$\omega(\zeta+\xi) + \zeta(\xi+\omega) + \xi(\omega+\zeta) = (\omega\zeta+\omega\xi) + (\zeta\xi+\zeta\omega) + (\xi\omega+\xi\zeta) = 2(\omega\zeta+\zeta\xi+\xi\omega).$$

The Cauchy-Schwarz inequality gives

$$\langle x, y\rangle^2 \leq \|x\|^2 \, \|y\|^2,$$

and hence

$$(\omega+\zeta+\xi)^2 \leq \left(\frac{\omega}{\zeta+\xi} + \frac{\zeta}{\xi+\omega} + \frac{\xi}{\omega+\zeta}\right) \cdot 2(\omega\zeta+\zeta\xi+\xi\omega).$$

Since $2(\omega\zeta+\zeta\xi+\xi\omega) > 0$, division yields

$$\frac{\omega}{\zeta+\xi} + \frac{\zeta}{\xi+\omega} + \frac{\xi}{\omega+\zeta} \geq \frac{(\omega+\zeta+\xi)^2}{2(\omega\zeta+\zeta\xi+\xi\omega)},$$

as required.

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12.6 Orthogonal Complements and Orthogonal Projections

DEFINITION 12.8

Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. For $S \subseteq V$, define the **orthogonal complement** of S by

$$S^{\perp} := \{\, v \in V : \ \forall s \in S,\ \langle v, s \rangle = 0 \,\}.$$

In particular, if $W \subseteq V$ is a subspace, then $W^{\perp}$ is the **orthogonal complement** of W.

DEFINITION 12.9

Let $U, W \subseteq V$ be subspaces. We say that U and W are **orthogonal** (written $U \perp W$) if

$$\forall u \in U \ \forall w \in W \qquad \langle u, w \rangle = 0.$$

PROPOSITION 12.6

Let $S \subseteq V$, and let $U, W \subseteq V$ be subspaces. The following statements hold:

1. $S^{\perp}$ is a subspace of V.
2. If $S \subseteq T \subseteq V$, then $T^{\perp} \subseteq S^{\perp}$.
3. $(\operatorname{span} S)^{\perp} = S^{\perp}$.
4. $W \cap W^{\perp} = \{0\}$.
5. $U \perp W \iff U \subseteq W^{\perp} \iff W \subseteq U^{\perp}$.
6. $(U + W)^{\perp} = U^{\perp} \cap W^{\perp}$.
7. If V is finite dimensional, then $(U \cap W)^{\perp} = U^{\perp} + W^{\perp}$.
8. If $U \perp W$, then $U + W = U \oplus W$.
9. $W \subseteq (W^{\perp})^{\perp}$. If V is finite dimensional, then $(W^{\perp})^{\perp} = W$.

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RECALL AND TOOLS

For all projection results below, assume that V is finite dimensional. In infinite dimensions, closedness of W alone is insufficient in a general inner product space: the projection theorem holds when V is complete in the induced norm, that is, a Hilbert space, and W is closed. Alternatively, it suffices that W be finite dimensional, without assuming completeness of V.

DEFINITION 12.10

Suppose V is finite dimensional and $W \subseteq V$ is a subspace. The **orthogonal projection onto** W is the map $P_W : V \to W$ defined by requiring that, for every $v \in V$, $P_W(v) \in W$ be the unique vector satisfying

$$v - P_W(v) \in W^{\perp}.$$

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PROPOSITION 12.7

Suppose V is finite dimensional and $W \subseteq V$ is a subspace. The following hold:

1. For every $v \in V$, there are unique $w \in W$ and $z \in W^\perp$ such that
$$v = w + z, \qquad P_W(v) = w.$$
In particular, $V = W \oplus W^\perp$.
2. P_W is linear.
3. $$P_W^2 = P_W, \qquad \mathrm{Im}(P_W) = W, \qquad \ker(P_W) = W^\perp.$$
4. For every $v \in V$,
$$P_W(v) \perp \big(v - P_W(v)\big), \qquad \|v\|^2 = \|P_W(v)\|^2 + \|v - P_W(v)\|^2.$$
5. For all $u, v \in V$,
$$\langle P_W(u), v\rangle = \langle u, P_W(v)\rangle.$$
6. For every $v \in V$, $\|P_W(v)\| \le \|v\|$.

RECALL AND TOOLS

If $\dim V < \infty$ and $\{e_1, \ldots, e_k\}$ is an orthonormal basis of W, then for every $v \in V$,
$$P_W(v) = \sum_{i=1}^{k} \langle v, e_i\rangle\, e_i,$$
and, in particular,
$$W^\perp = \{\, v \in V : \ \langle v, e_i\rangle = 0 \ \forall i = 1, \ldots, k \,\}.$$

PROPOSITION 12.8

If $\dim V < \infty$ and $W \subseteq V$ is a subspace, then
$$\dim W + \dim W^\perp = \dim V.$$

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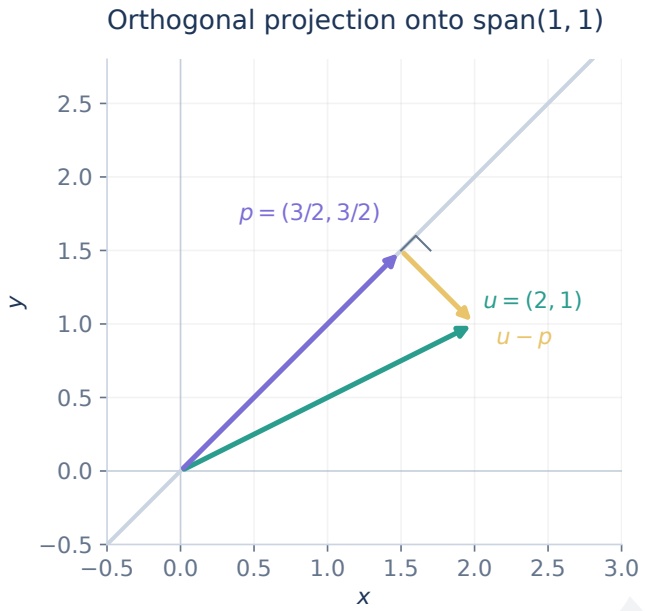


Figure 12.1. Orthogonal projection decomposes a vector into a component in the subspace and a perpendicular residual. The two components satisfy the Pythagorean identity.

EXERCISE 12.8

Let $y = (-2, 9)^T$, $u = (2, 1)^T$, and $W = \text{span}\{u\}$. For the weighted inner product $\langle x, z\rangle_M = x^T M z$, where $M = \text{diag}(1, 4)$, please compute $P_W^M(y)$ and please check that the residual is orthogonal to W. Please compare with the projection $P_W^I(y)$ for the standard inner product.

WORKED SOLUTION

M is symmetric and positive definite, since $x^T M x = x_1^2 + 4x_2^2 > 0$ when $x \neq 0$. The coefficient of the weighted projection is

$$t_M = \frac{y^T M u}{u^T M u} = \frac{(-2)\cdot 2 + 4 \cdot 9 \cdot 1}{2^2 + 4 \cdot 1^2} = \frac{-4 + 36}{8} = 4.$$

Thus $P_W^M(y) = 4u = (8, 4)^T$, and the residual is $r_M = (-10, 5)^T$. Check:

$$r_M^T M u = (-10)\cdot 2 + 4 \cdot 5 \cdot 1 = -20 + 20 = 0.$$

For the standard inner product,

$$t_I = \frac{y^T u}{u^T u} = \frac{-4 + 9}{4 + 1} = 1, \qquad P_W^I(y) = u = (2, 1)^T.$$

The residual is $r_I = (-4, 8)^T$; indeed, $r_I^T u = -4 \cdot 2 + 8 \cdot 1 = 0$. In both cases, the projection lies in W and the residual is orthogonal to it for the relevant inner product. Changing the inner product changes the projection; here both projections have integer coordinates.

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12.7 Linear Maps on Inner Product Spaces

EXERCISE 12.9

Let $(V, \langle\cdot,\cdot\rangle)$ be a real inner product space of dimension $n \geq 3$. Let $v, w \in V$ be orthonormal: that is, $\langle v, v\rangle = \langle w, w\rangle = 1$ and $\langle v, w\rangle = 0$. Define the linear map $T : V \to V$ by

$$T(u) = \langle u, v\rangle\, w + \langle u, w\rangle\, v.$$

1. Please find all eigenvalues of T, all eigenvectors, and the characteristic polynomial of T.
2. Please prove that $\|T(u)\| \leq \|u\|$ for every $u \in V$, and please find all u for which equality holds.

WORKED SOLUTION

1. Start with the eigenvalue equation and compare components using an orthogonal decomposition:

   $$T(u) = \lambda u.$$

   Decompose each $u \in V$ using the orthogonal direct sum

   $$V = \text{span}\{v, w\} \oplus (\text{span}\{v, w\})^{\perp}.$$

   Here $\{v, w\}^{\perp}$ denotes the orthogonal complement of $\text{span}\{v, w\}$. Thus there are unique $\alpha, \beta \in \mathbb{R}$ and $y \in \{v, w\}^{\perp}$ such that

   $$u = \alpha v + \beta w + y.$$

   Now compute $T(u)$: since $y \perp v, w$, we have $\langle y, v\rangle = \langle y, w\rangle = 0$, so

   $$\langle u, v\rangle = \langle \alpha v + \beta w + y,\ v\rangle = \alpha, \qquad \langle u, w\rangle = \langle \alpha v + \beta w + y,\ w\rangle = \beta.$$

   Hence

   $$T(u) = \alpha w + \beta v.$$

   Substitute into the eigenvalue equation $T(u) = \lambda u$ to obtain

   $$\alpha w + \beta v = \lambda(\alpha v + \beta w + y).$$

   Compare components in the direct sum $\text{span}\{v, w\} \oplus \{v, w\}^{\perp}$. The component in $\{v, w\}^{\perp}$ on the left is zero, so necessarily

   $$\lambda y = 0.$$

   Consequently:

   - If $\lambda \neq 0$, then $y = 0$, so $u \in \text{span}\{v, w\}$.
   - If $\lambda = 0$, then $T(u) = 0$, and $\alpha w + \beta v = 0$ gives $\alpha = \beta = 0$, so $u = y \in \{v, w\}^{\perp}$.

   Now consider $\lambda \neq 0$. Then $u = \alpha v + \beta w$ and $y = 0$. The eigenvalue equation becomes

   $$\alpha w + \beta v = \lambda(\alpha v + \beta w).$$

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EXERCISE 12.9 CONTINUED

Comparing coefficients of v and w gives the system

$$\beta = \lambda\alpha, \qquad \alpha = \lambda\beta.$$

Substitution gives $\alpha = \lambda(\lambda\alpha) = \lambda^2\alpha$, so

$$(\lambda^2 - 1)\alpha = 0.$$

If $\alpha = 0$, then $\beta = \lambda\alpha = 0$ as well, giving $u = 0$, contrary to the definition of an eigenvector. Thus $\alpha \neq 0$, and

$$\lambda^2 - 1 = 0 \quad \Longleftrightarrow \quad \lambda \in \{1, -1\}.$$

Now determine the eigenvectors:

- For $\lambda = 1$, $\beta = \alpha$, so $u = \alpha(v + w)$; hence
$$V_1 = \text{span}\{v + w\}.$$
- For $\lambda = -1$, $\beta = -\alpha$, so $u = \alpha(v - w)$; hence
$$V_{-1} = \text{span}\{v - w\}.$$
- For $\lambda = 0$, we obtained $u \in \{v, w\}^\perp$, so
$$V_0 = \{v, w\}^\perp.$$

In particular, there are no other eigenvalues, because the eigenvalue equation forced $\lambda \in \{0, 1, -1\}$.

To compute the characteristic polynomial, choose an orthonormal basis of V of the form

$$B = (v, w, e_3, \ldots, e_n),$$

where $e_3, \ldots, e_n$ is an orthonormal basis of $\{v, w\}^\perp$. Then

$$T(v) = w, \qquad T(w) = v, \qquad T(e_i) = 0 \ \ (i = 3, \ldots, n).$$

The matrix representing T relative to B is therefore

$$[T]_B = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix}.$$

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EXERCISE 12.9 CONTINUED

Hence its characteristic polynomial is

$$p_T(\lambda) = \det(\lambda I - [T]_B) = \det\begin{pmatrix} \lambda & -1 & 0 & \cdots & 0 \\ -1 & \lambda & 0 & \cdots & 0 \\ 0 & 0 & \lambda & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda \end{pmatrix} = (\lambda^2 - 1)\lambda^{n-2}.$$

2. Set $\alpha := \langle u, v\rangle$ and $\beta := \langle u, w\rangle$, and define

$$s := \alpha v + \beta w \in \text{span}\{v, w\}.$$

Since v, w are orthonormal,

$$\|s\|^2 = \langle s, s\rangle = \alpha^2 + \beta^2 = \langle u, v\rangle^2 + \langle u, w\rangle^2 = \|T(u)\|^2.$$

Also,

$$\langle u, s\rangle = \alpha\langle u, v\rangle + \beta\langle u, w\rangle = \alpha^2 + \beta^2 = \|s\|^2.$$

By the Cauchy-Schwarz inequality,

$$|\langle u, s\rangle| \le \|u\|\,\|s\|.$$

Since $|\langle u, s\rangle| = \|s\|^2$, we obtain

$$\|s\|^2 \le \|u\|\,\|s\|.$$

If $s = 0$, then $\|T(u)\| = \|s\| = 0 \le \|u\|$. If $s \neq 0$, divide by $\|s\|$ to obtain

$$\|T(u)\| = \|s\| \le \|u\|.$$

To check equality, write $u = s + y$ with $y \in \{v, w\}^\perp$. By the Pythagorean identity,

$$\|u\|^2 = \|s\|^2 + \|y\|^2 = \|T(u)\|^2 + \|y\|^2.$$

Thus $\|T(u)\| = \|u\|$ if and only if $y = 0$, or equivalently $u \in \text{span}\{v, w\}$. This argument also includes $s = 0$, in which case equality requires $u = 0$.

EXERCISE 12.10

Let $n \ge 3$ be a natural number and let $a \in \mathbb{R}$. Define $A_n(a) \in M_n(\mathbb{R})$ by

$$(A_n(a))_{ij} = \begin{cases} 1, & i = j, \\ a, & i \neq j. \end{cases}$$

Recall the definition: a Hermitian matrix $H \in M_n(\mathbb{C})$ is positive semidefinite if $x^*Hx \ge 0$ for every $x \in \mathbb{C}^n$, and positive definite if $x^*Hx > 0$ for every $x \neq 0$.

1. For every a, please find all eigenvalues of $A_n(a)$ and their multiplicities.
2. Please determine for which values of a the matrix $A_n(a)$ is positive definite, and for which values it is positive semidefinite.

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EXERCISE 12.10 CONTINUED

3. Please find $\operatorname{rank}(A_n(a))$ and $\ker(A_n(a))$ in the two boundary cases where $A_n(a)$ is positive semidefinite but not positive definite.

In addition, let $B \in M_n(\mathbb{C})$ and please write $B^* = \overline{B}^T$. Define

$$H := B^*B + I.$$

(4) Please prove that H is positive definite and hence invertible.

(5) Please prove that, for every $x \in \mathbb{C}^n$,

$$x^*H^{-1}x \leq \|x\|^2,$$

and also prove that

$$\det(H) \geq 1,$$

with equality if and only if $B = 0$.

WORKED SOLUTION

Write $e = (1, 1, \ldots, 1)^T \in \mathbb{R}^n$. We have the decomposition

$$A_n(a) = (1-a)I + a\,ee^T,$$

because every entry of ee^T equals 1.

1. Note that $ee^Te = (e^Te)\,e = n\,e$, so

$$A_n(a)e = (1-a)e + a\,ee^Te = (1-a)e + an\,e = \big(1 + (n-1)a\big)e,$$

Thus e is an eigenvector with eigenvalue $\lambda_1 = 1 + (n-1)a$.
Now let $x \in \mathbb{R}^n$ satisfy $e^Tx = 0$. Then $ee^Tx = e(e^Tx) = 0$, so

$$A_n(a)x = (1-a)x + a \cdot 0 = (1-a)x.$$

Consequently, every $x \in e^\perp \setminus \{0\}$ is an eigenvector with eigenvalue $\lambda_2 = 1 - a$. Since $\mathbb{R}^n = \operatorname{span}\{e\} \oplus e^\perp$, these are all eigenvalues. If $a \neq 0$, then $\lambda_1 \neq \lambda_2$, so

$$\begin{aligned} &\lambda_1 = 1 + (n-1)a && \text{with algebraic and geometric multiplicity } 1, \\ &\lambda_2 = 1 - a && \text{with algebraic and geometric multiplicity } n-1. \end{aligned}$$

If $a = 0$, then $A_n(0) = I$, so its only eigenvalue is 1, with algebraic and geometric multiplicity n.

2. The matrix $A_n(a)$ is real symmetric and hence Hermitian. It is therefore positive definite exactly when all its eigenvalues are strictly positive, and positive semidefinite exactly when all its eigenvalues are nonnegative. Thus:

$$A_n(a) \succ 0 \iff \begin{cases} 1-a > 0, \\ 1 + (n-1)a > 0, \end{cases} \iff -\frac{1}{n-1} < a < 1,$$

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EXERCISE 12.10 CONTINUED

and

$$A_n(a) \succeq 0 \iff \begin{cases} 1-a \geq 0, \\ 1+(n-1)a \geq 0, \end{cases} \iff -\frac{1}{n-1} \leq a \leq 1.$$

3. If $a = 1$, then $\lambda_2 = 0$ has multiplicity $n-1$, and $\lambda_1 = n$. Hence

$$\operatorname{rank}(A_n(1)) = 1, \qquad \ker(A_n(1)) = e^{\perp} = \{x \in \mathbb{R}^n : \sum_{i=1}^{n} x_i = 0\}.$$

If $a = -\frac{1}{n-1}$, then $\lambda_1 = 0$ has multiplicity 1, and $\lambda_2 = 1 - a = \frac{n}{n-1} > 0$ has multiplicity $n-1$. Hence

$$\operatorname{rank}\Big(A_n\big(-\tfrac{1}{n-1}\big)\Big) = n-1, \qquad \ker\Big(A_n\big(-\tfrac{1}{n-1}\big)\Big) = \operatorname{span}\{e\}.$$

Now turn to $H = B^*B + I$.

(4) Let $x \in \mathbb{C}^n$. Then

$$x^*Hx = x^*(B^*B)x + x^*Ix = \|Bx\|^2 + \|x\|^2.$$

Hence $x^*Hx \geq 0$ for every x, and $x^*Hx > 0$ for every $x \neq 0$, since $\|x\|^2 > 0$. Thus $H \succ 0$, so H is invertible.

(5) From $H = I + B^*B$, we obtain $H - I = B^*B \succeq 0$; thus $x^*Hx \geq x^*Ix = \|x\|^2$ for every x. Let $\lambda_{\min}$ be the smallest eigenvalue of H. The preceding inequality gives $\lambda_{\min} \geq 1$. Every eigenvalue of H^{-1} is therefore at most 1, so for every x,

$$x^*H^{-1}x \leq \|x\|^2.$$

For the determinant inequality, write the eigenvalues of H as $\lambda_1, \ldots, \lambda_n$. Since $H \succ 0$, we have $\lambda_i > 0$; since $\lambda_{\min} \geq 1$, in fact $\lambda_i \geq 1$ for every i. Hence

$$\det(H) = \prod_{i=1}^{n} \lambda_i \geq 1.$$

Equality holds if and only if $\lambda_i = 1$ for every i: equivalently, $H = I$, or $B^*B = 0$. Finally, $B^*B = 0$ implies $\|Bx\|^2 = x^*B^*Bx = 0$ for every x. Thus $Bx = 0$ for all x, so $B = 0$.

EXERCISE 12.11 EXAM QUESTION

1. Work in $M_{3\times 4}(\mathbb{R})$. Define

$$C = \operatorname{diag}(1,0,1), \qquad D = \operatorname{diag}(1,0,0,1),$$

and

$$W = \{X \in M_{3\times 4}(\mathbb{R}) : CXD = 0\}.$$

Please prove that W is a subspace of $M_{3\times 4}(\mathbb{R})$.

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EXERCISE 12.11 EXAM QUESTION CONTINUED

2. On $M_{3\times4}(\mathbb{R})$, define

$$\langle X, Y\rangle = \operatorname{tr}(X^T Q Y), \qquad Q = \operatorname{diag}(1,2,3).$$

Please prove that this is an inner product. In particular, please verify linearity in the second argument.

3. Now work in $\mathbb{R}^4$ with the standard inner product. Let

$$M = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \qquad W = N(M), \qquad v = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}.$$

Please compute the orthogonal projection $P_W(v)$.

WORKED SOLUTION

1. Define $L : M_{3\times4}(\mathbb{R}) \longrightarrow M_{3\times4}(\mathbb{R})$ by $L(X) = CXD$. For all $\alpha, \beta \in \mathbb{R}$ and all X_1, X_2,

$$\begin{aligned} L(\alpha X_1 + \beta X_2) &= C(\alpha X_1 + \beta X_2)D \\ &= \alpha C X_1 D + \beta C X_2 D \\ &= \alpha L(X_1) + \beta L(X_2). \end{aligned}$$

Thus L is linear and $W = \ker L$, so W is a subspace. The condition can also be read directly: multiplication by C zeros the second row, and multiplication by D zeros the second and third columns. Hence

$$CXD = 0 \quad \Longleftrightarrow \quad x_{11} = x_{14} = x_{31} = x_{34} = 0.$$

These are homogeneous linear conditions on the entries of X.

2. Write $Q = \operatorname{diag}(q_1, q_2, q_3)$, where $(q_1, q_2, q_3) = (1, 2, 3)$. If $X = (x_{ij})$ and $Y = (y_{ij})$, computation of the diagonal entries gives

$$\begin{aligned} (X^T Q Y)_{jj} &= \sum_{i=1}^{3} x_{ij} q_i y_{ij}, \\ \langle X, Y\rangle &= \sum_{j=1}^{4} \sum_{i=1}^{3} q_i x_{ij} y_{ij}. \end{aligned}$$

For all $\alpha, \beta \in \mathbb{R}$,

$$\begin{aligned} \langle X, \alpha Y + \beta Z\rangle &= \sum_{j=1}^{4} \sum_{i=1}^{3} q_i x_{ij} (\alpha y_{ij} + \beta z_{ij}) \\ &= \alpha \langle X, Y\rangle + \beta \langle X, Z\rangle. \end{aligned}$$

This proves linearity in the second argument. Since all entries are real, the same calculation also proves linearity in the first argument. In addition,

$$\langle X, Y\rangle = \sum_{j=1}^{4} \sum_{i=1}^{3} q_i y_{ij} x_{ij} = \langle Y, X\rangle,$$

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EXERCISE 12.11 EXAM QUESTION CONTINUED

so symmetry holds. Finally,

$$\langle X, X\rangle = \sum_{j=1}^{4}(x_{1j}^2 + 2x_{2j}^2 + 3x_{3j}^2) \geq 0.$$

Equality holds if and only if all entries x_{ij} vanish, or equivalently $X = 0$. We have proved linearity, symmetry, and positive definiteness; thus this is a real inner product.

3. Solve $Mx = 0$:

$$x_1 + x_3 = 0, \qquad x_2 + x_4 = 0.$$

Thus

$$W = \text{span}\{w_1, w_2\}, \qquad w_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}, \quad w_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}.$$

The two vectors are linearly independent and orthogonal, and

$$\begin{aligned} \langle w_1, w_2\rangle &= 0, & \|w_1\|^2 &= \|w_2\|^2 = 2, \\ \langle v, w_1\rangle &= 1 - 3 = -2, & \langle v, w_2\rangle &= 2 - 4 = -2. \end{aligned}$$

The projection formula for an orthogonal basis therefore gives

$$\begin{aligned} P_W(v) &= \frac{\langle v, w_1\rangle}{\|w_1\|^2} w_1 + \frac{\langle v, w_2\rangle}{\|w_2\|^2} w_2 \\ &= -w_1 - w_2 = \begin{pmatrix} -1 \\ -1 \\ 1 \\ 1 \end{pmatrix}. \end{aligned}$$

As a check, $MP_W(v) = 0$, whereas

$$v - P_W(v) = \begin{pmatrix} 2 \\ 3 \\ 2 \\ 3 \end{pmatrix}$$

is orthogonal to both w_1 and w_2, since $2 - 2 = 0$ and $3 - 3 = 0$. The projection indeed belongs to W, and the residual is orthogonal to all of W.

Additional practice

EXERCISE 12.12 FOUNDATION

On $\mathbb{R}^2$, consider the inner product $\langle x, y\rangle_G = 2x_1y_1 + x_2y_2$. Let $u = (1, 1)^T$, $v = (2, 0)^T$, and $W = \text{span}\{u\}$.

1. Please compute $p = P_W(v)$ and the residual $r = v - p$.
2. Please verify both conditions characterizing the projection and the identity $\|v\|_G^2 =$

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EXERCISE 12.12 FOUNDATION CONTINUED

$\|p\|_G^2 + \|r\|_G^2$.

WORKED SOLUTION

1. The matrix $G = \mathrm{diag}(2,1)$ is symmetric and positive definite, since $2x_1^2 + x_2^2 > 0$ for every $x \neq 0$. We have

$$\langle v, u\rangle_G = 4, \qquad \langle u, u\rangle_G = 3,$$

so

$$p = \frac{4}{3}u = \begin{pmatrix} 4/3 \\ 4/3 \end{pmatrix}, \qquad r = \begin{pmatrix} 2/3 \\ -4/3 \end{pmatrix}.$$

2. The expression $p = \frac{4}{3}u$ shows that $p \in W$. Also,

$$\langle r, u\rangle_G = 2 \cdot \frac{2}{3} - \frac{4}{3} = 0,$$

so $r \perp W$. These two conditions guarantee that p is the required projection. Finally,

$$\|v\|_G^2 = 8, \qquad \|p\|_G^2 = 2 \cdot \frac{16}{9} + \frac{16}{9} = \frac{16}{3}, \qquad \|r\|_G^2 = 2 \cdot \frac{4}{9} + \frac{16}{9} = \frac{8}{3}.$$

Indeed, $8 = \frac{16}{3} + \frac{8}{3}$.

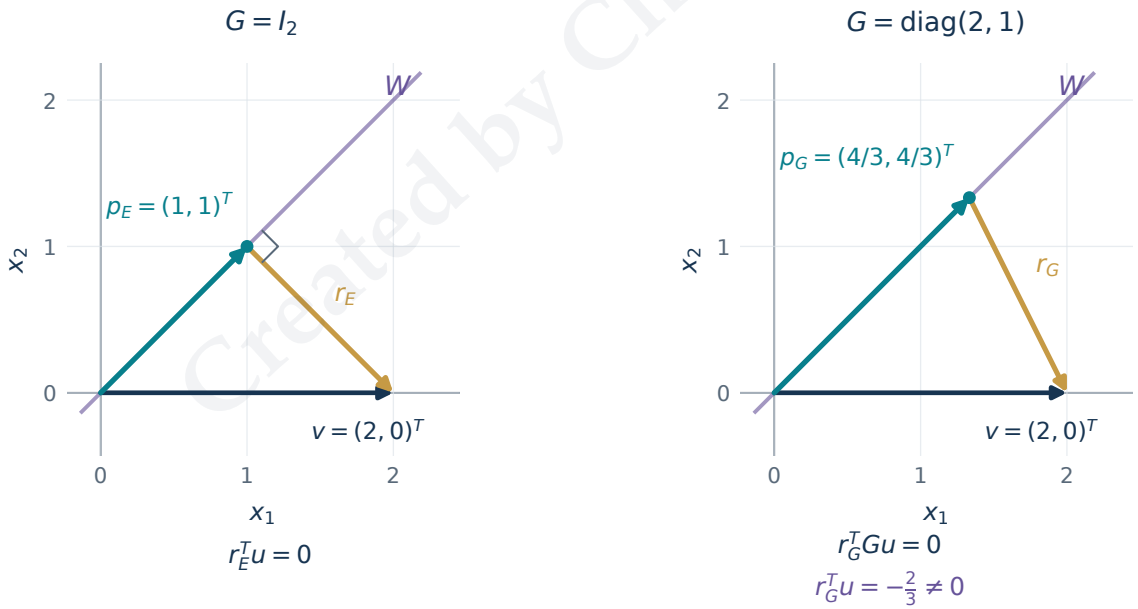


Figure 12.2. Both panels use $v = (2,0)^T$ and $W = \mathrm{span}\{u\}$, where $u = (1,1)^T$. On the left, the standard inner product gives $p_E = (1,1)^T$; on the right, $\langle x, y\rangle_G = x^T G y$ with $G = \mathrm{diag}(2,1)$ gives $p_G = (4/3, 4/3)^T$. Gold arrows are the residuals. On the right, $r_G^T G u = 0$, whereas $r_G^T u = -2/3 \neq 0$: orthogonality is with respect to the weighted inner product, so no Euclidean right-angle marker appears there.

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EXERCISE 12.13 CHALLENGE

Let V be a finite-dimensional inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and let $U, W \leq V$. Please write $P = P_U$ and $Q = P_W$, viewed as maps from V to itself.

1. Please prove that PQ is the orthogonal projection onto $U \cap W$ if and only if $PQ = QP$.
2. In $\mathbb{R}^2$ with its standard inner product, take $U = \text{span}\{(1,0)^T\}$ and $W = \text{span}\{(1,1)^T\}$. Please compute PQ and please check whether it is an orthogonal projection.

WORKED SOLUTION

1. Use $\langle Px, y\rangle = \langle x, Py\rangle$, valid for every orthogonal projection, and the corresponding identity for Q. First suppose $PQ = QP$, and set $R = PQ$. For each $x \in V$, the vector Rx belongs to U; since $Rx = QPx$, it also belongs to W. For every $z \in U \cap W$, we have $Pz = Qz = z$, so $Rz = z$. Furthermore,

$$\langle Rx, z\rangle = \langle PQx, z\rangle = \langle Qx, Pz\rangle = \langle x, QPz\rangle = \langle x, z\rangle.$$

Thus $x - Rx \perp U \cap W$. The two defining conditions for projection show that $R = P_{U\cap W}$. Conversely, suppose $R = PQ = P_{U\cap W}$. For all $x, y \in V$,

$$\langle x, QPy\rangle = \langle PQx, y\rangle = \langle Rx, y\rangle = \langle x, Ry\rangle = \langle x, PQy\rangle.$$

Hence $\langle x, (QP - PQ)y\rangle = 0$ for all x, y. For fixed y, choose $x = (QP - PQ)y$; positive definiteness gives $(QP - PQ)y = 0$. Therefore $QP = PQ$.

2. Here

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad Q = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad PQ = \begin{pmatrix} 1/2 & 1/2 \\ 0 & 0 \end{pmatrix}.$$

However,

$$(PQ)^2 = \begin{pmatrix} 1/4 & 1/4 \\ 0 & 0 \end{pmatrix} \neq PQ.$$

Every orthogonal projection satisfies $R^2 = R$, so PQ is not an orthogonal projection. Indeed, $U \cap W = \{0\}$, whose projection is the zero map, whereas $PQ \neq 0$.

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EXERCISE 12.14 CHALLENGE

Let $a_1, \dots, a_m$ lie in an n-dimensional real inner product space, where $n \geq 1$, and assume $\|a_i - a_j\| = 1$ for all $i \neq j$. Please prove that $m \leq n+1$. Please construct an example attaining the bound in an n-dimensional subspace of $\mathbb{R}^{n+1}$.

WORKED SOLUTION

The case $m = 1$ is immediate. Otherwise put $v_i = a_i - a_m$ for $i < m$. Then $\|v_i\| = 1$, and for $i \neq j$,

$$1 = \|v_i - v_j\|^2 = 2 - 2\langle v_i, v_j \rangle.$$

Thus $\langle v_i, v_j \rangle = 1/2$. For real scalars c_i,

$$\left\| \sum_i c_i v_i \right\|^2 = \frac{1}{2} \sum_i c_i^2 + \frac{1}{2} \left(\sum_i c_i \right)^2 .$$

A zero combination therefore has every $c_i = 0$. The $m - 1$ vectors are independent, giving $m - 1 \leq n$. For an example, let $\mathbf{1} = (1, \dots, 1)^T \in \mathbb{R}^{n+1}$ and take

$$H = \{x : x_1 + \cdots + x_{n+1} = 0\}, \qquad a_j = \frac{1}{\sqrt{2}} \left(e_j - \frac{1}{n+1} \mathbf{1} \right).$$

The space H has dimension n, and all $n+1$ points lie in it. For $i \neq j$, $\|a_i - a_j\|^2 = \|e_i - e_j\|^2/2 = 1$, so the bound is attained.

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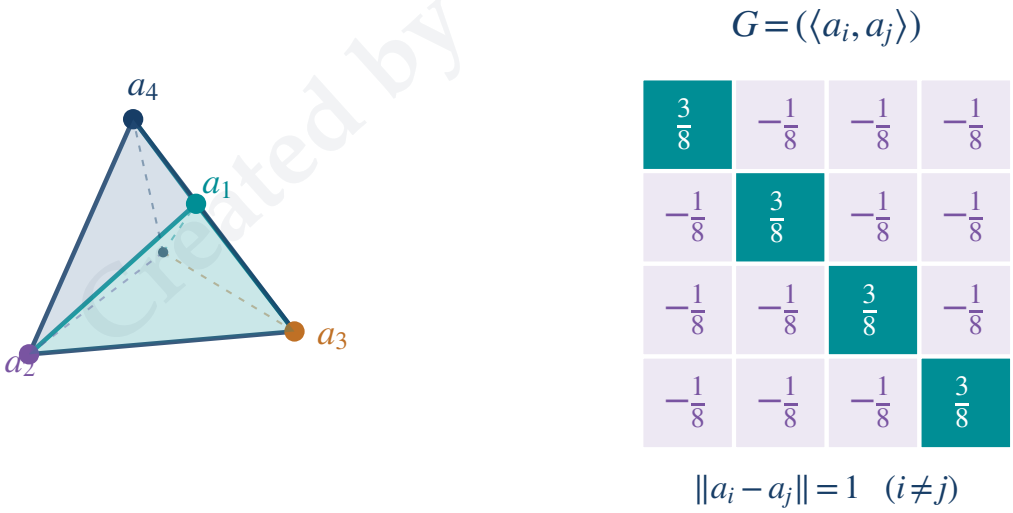


Figure 12.3. In dimension 3, four points can have all pairwise distances equal. The matrix records inner products of the vertices after their centre is placed at zero.

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EXERCISE 12.15 CHALLENGE

Let V be a complex inner product space, with the inner product linear in its first argument. Please prove that

$$4\langle u, v\rangle = \|u+v\|^2 - \|u-v\|^2 + i\big(\|u+iv\|^2 - \|u-iv\|^2\big).$$

Let $T : V \to V$ be complex-linear and satisfy $\|Tx\| = \|x\|$ for every x. Please deduce that $\langle Tu, Tv\rangle = \langle u, v\rangle$ for every u, v.

WORKED SOLUTION

Put $z = \langle u, v\rangle$. Expanding gives

$$\|u+v\|^2 = \|u\|^2 + \|v\|^2 + z + \overline{z}, \qquad \|u-v\|^2 = \|u\|^2 + \|v\|^2 - z - \overline{z}.$$

Their difference is $4\,\mathrm{Re}\, z$. Conjugate-linearity in the second argument gives $\langle u, iv\rangle = -iz$ and $\langle iv, u\rangle = i\overline{z}$. Consequently

$$\|u+iv\|^2 = \|u\|^2 + \|v\|^2 - iz + i\overline{z},$$
$$\|u-iv\|^2 = \|u\|^2 + \|v\|^2 + iz - i\overline{z}.$$

Their difference is $4\,\mathrm{Im}\, z$, proving the formula. Apply it to Tu, Tv. Complex linearity gives $Tu \pm Tv = T(u \pm v)$ and $Tu \pm iTv = T(u \pm iv)$. Norm preservation replaces all four norms by the original ones, so the inner product is preserved as well.

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